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An evaluation and optimization of nutrition, environmental footprint, and food waste in Italian primary school menus: a case study

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Abstract

Background Adherence of young people to the Mediterranean diet is increasingly concerning, as their dietary habits present several critical challenges. It is crucial to explore how educational institutions can intervene, particularly through school canteens. These are important foundations in shaping students' eating habits with the potential to promote a healthy and sustainable lifestyle. A balanced diet can also contribute significantly to sustainability, reducing the environmental impact of food choices. In this context, this study aims to optimize the primary school canteen menu in Camerino (MC), Italy, enhancing nutritional quality promoting healthier practices and reducing environmental impact.

Methods This case study involves the nutritional assessment analysis of the canteen menus (the baseline and the revised one) to evaluate energy and macronutrient intake. Environmental impact, referred to as carbon and water footprints, was assessed. Food waste data were collected by waste weighing.

Results The school canteen menu was changed from a 4-week to an 8-week cycle to introduce greater variety while maintaining Mediterranean Diet principles adherence. Frequencies were modified by increasing legume frequency from about once to 2 times per school week and reducing meat consumption, decreasing white meat to once per school week and red meat from twice per school week to three times per 8 school week. Processed meats were eliminated. Nutritional analysis showed no significant differences in most values, except for fibre content increased from 7.8 ± 0.6 to 8.9 ± 1.2 g per meal. The revised menu emphasized seasonality, local sourcing, and organic products, leading to a reduction in the environmental impact. The carbon footprint significantly decreased from 5.2 ± 1.6 to 3.7 ± 0.9 kg CO₂eq per meal, and the water footprint was reduced from 5176.0 ± 738.9 to 4608 ± 708.8 L per meal. Food waste levels were recorded at 24.4% for the first course, 26.0% for the second course, and 31.4% for side dishes, with the highest waste occurring in vegetable-based side dishes.

Conclusions Optimization of the menus plays a key role in encouraging sustainable and healthier eating behaviours. Therefore, schools, institutions and municipalities have a crucial strategic role by giving educational support.

Keywords School Menu, Carbon footprint, Water footprint, Nutritional assessment, Food waste

Background

The increase in childhood overweight and obesity represents one of the most critical global public health challenges today [1].

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In Italy, 19% of children are overweight and 9.8% are obese, including 2.6% with severe obesity, placing Italy among the Mediterranean countries with the highest rate of childhood overweight [2, 3]. Childhood obesity increase the risk of developing non-communicable diseases (NCDs), such as diabetes and cardiovascular diseases and hepatic complications in adult life [4] and are associated with obesity over the life course [5]. Unhealthy eating habits and low levels of physical activity are widely recognized as contributing factors [6] and national findings show a growing divergence between children's dietary habits and nutritional recommendations [2] highlighting the need of promoting healthy dietary habits and reduce sedentary behaviour [7].

In this context is repeated two times, schools represent strategic environments for the promotion of health and sustainability from early childhood. School canteen has a pivotal role not only in contributing to the daily nutritional intake by contributing to approximately 35% of their recommended daily nutritional requirements but also act as platforms for food literacy and nutrition education programs [8–11].

In Italy, the Ministry of Health has developed a specific document containing guidelines for school canteen emphasizing the importance of the healthy eating plate and promoting. The consumption of fruits, vegetables, and pulses and the respect of local tradition and seasonality, all in line with Mediterranean Diet principles. The adoption of the Mediterranean Diet (MD) within school meal programs is strongly encouraged, aiming to enhance students' nutritional status while concurrently reducing the environmental impact of food choices [12–16].

Besides social and health role, school canteen can promote environmental sustainability by encouraging sustainable food production and consumption, one of the Sustainable Development Goals (SDGs) of the Agenda 2030 [17]. Food production and food system are the main responsible for environmental change, biodiversity loss and natural resource degradation [18, 19]. Diets rich in energy content, added sugar, saturated fats, processed foods and red meats are associated with increased greenhouse gas (GHG) emissions, land use and water use. Thus, dietary composition influences not only human health but also environmental outcomes [18].

In this context, sustainable diets are linked to the concept of addressing health and environmental concerns related to food production and consumption [19].

In order to reach the Agenda 2030 goals and promote food sustainability, the EAT Lancet Commission recommend a healthy reference diet, based on a low to moderate amount of fish and poultry and a strict reduction of red meat consumption, supporting a shift towards more plant-based diets [19].

In recent years, increasing attention has been focused on the integration of environmental sustainability into catering services [20]. In this context, the *Minimum Environmental Criteria* (Criteri Ambientali Minimi, CAM), a document containing guidelines to reduce the impact on the environment defined in the Italian legislative decree of the 10 th March 2020 [15], encourage the inclusion of organic and locally sourced foods, the reduction of packaging and food waste, and improved supply chain planning and waste monitoring.

Since it is estimated that waste in school catering ranges from 35–40%, the reduction of food waste is crucial to minimize the impact on the planet [12, 14, 15]. For this, the CAM proposes targeted actions to minimize food waste, and reduce the environmental impact of the canteen by optimizing the process of food preparation, conservation, and administration. CAM specifically advocate for the use of seasonal, locally sourced, and organic food products, specifying that at least 50% by weight of fruits, vegetables, legumes, cereals, and bovine meat should be organic. CAM also encourages the consumption of low cost-effective and low input demanding and polluting protein sources, such as legumes, less commonly consumed fish species, and alternative meat cuts. Additionally, it promotes the use of recipes that incorporate typically discarded, yet edible, parts of fruits and vegetables to reduce food waste while emphasizing the reduction of products with excessive packaging, aiming to decrease the reliance on plastic materials [15].

Food waste reduction has become a goal worldwide to decrease the impact on the environment, especially resource depletion and global warming as also addressed in the United Nations' Agenda 2030 [17]. Globally, 17% of the food produced is wasted due to the retail or consumption level [21]. It is estimated that food losses and waste represent 8–10% of the world's GHG emissions, being the third biggest source of GHG [22].

In Italy, Eurostat reports 139 kg per capita of food including production sites, processing and manufacture, retail and distribution, restaurants and food services and households generated in 2022 (*Eurostat*, 2022).

Istat report 2023 underlines that the 53% of children aged 3–5 years and 41.3% aged 6–10 have lunch in the school canteen (*Istat*, 2023). Given that, the school canteen represents a key setting for both food waste prevention and education [23, 24].

Thus, the study of menu planning plays a critical role in promoting both individual health and environmental sustainability. Thoughtfully designed menus can significantly influence dietary choices and support optimal health outcomes for individuals, with also implications for environmental health.

In this context, the present study aims to revise the primary school canteen of the Camerino municipality (MC), Italy, by optimizing both its nutritional quality and environmental impact. The nutritional profile and environmental consequences of the meals served will be evaluated, with a particular focus on their carbon and water footprints. Additionally, the study will assess the food waste generated following the implementation of the revised menu. Thus, the goal is to enhance the menu from both a nutritional and environmental perspective, looking at the school canteen menu as a tool for nutrition education and promoting healthier lifestyles among students.

Methods

This case study took place in the Camerino (MC, Italy) primary school canteen from October 2022 to May 2023.

As first step a revision of the autumn–winter canteen menu was carried out developing a new general menu both for autumn/winter and spring/summer (with specific indications about the seasonality of plant-based products) taking into account the National guidelines for the collective school catering [12, 14], the Guide to the construction of the school canteen menu [25], and the CAM: Minimum Environmental Criteria for Collective Catering Service and Food Supply [15]. The revision took place together with municipality and reference nutritionist.

Nutritional assessment

Both menus (the baseline and the revised one) were analyzed using WinFood® by Medimatica to evaluate the energy and macronutrients contents. Since the baseline menu was structured as 4-week cycle menu while the revised one as a 8-week cycle menu in order to increase foods variability, the evaluation and comparison were carried out over the first 4 weeks. Specifically, each meal was assessed, considering the recommended portion sizes for the primary school lunch menu specified in the guide to constructing the school canteen menu [25]. The meal assessment evaluated each ingredient in the recipe, assuming that the meal was entirely consumed by the student.

The parameters considered were energy content (kcal), carbohydrate content (g and percentage of total meal energy content), simple carbohydrates intake (g), total protein intake (g and percentage of total meal energy content), animal protein and vegetable protein content (g and percentage of total proteins), total lipid intake (g and percentage of total meal energy content), saturated fat intake (g and percentage of total meal energy content) and fiber intake (g per meal and g per 1000 kcal).

Assessment of environmental impact

The environmental impact of both menus was evaluated based on two factors: carbon footprint (CF) and water footprint (WF). The Barilla Center for Food and Nutrition's database was used. This database is a multi-level dataset of CF and WF for food items and includes footprint data gathered from global surveys [26]. The CF reflects the total greenhouse gases emitted during a food product's life cycle, from production to distribution, indicating its contribution to global GHG emissions. The WF quantifies the amount of water used and polluted across all production stages, serving as a reliable measure for comparing the environmental impacts of different food types and diets [26]. For each food item, the database includes kg CO_{2eq}/kg for CF and liters of water/kg for WF.

In the assessment of the environmental impact of both menus, every ingredient of each dish was individually assessed according to portion sizes for primary schools specified in the guidelines [25].

For each food composing each meal, the two indicators were estimated through the Barilla Center for Food and Nutrition's database, a multilevel open-access database that provides carbon and water footprint values of food commodities based on standardized life cycle assessment approach [26]. Generalizable estimates were used suitable for comparisons across meals and food groups; selected values do not account for variability in food sourcing, such as local versus non-local origin thus no regional disaggregation was applied. The database has some limitations, including a Eurocentric bias in carbon footprint data, limited representativeness for certain items due to scarce or heterogeneous sources, and the exclusion of post-market phases such as cooking and waste disposal from its system boundaries [26].

For each food item included in the school meals, the two environmental indicators, carbon footprint (CF) and water footprint (WF), were estimated in accordance with the methodological guidelines provided [26]. The environmental impact of both the baseline and revised menus was then assessed by independently considering the CF values and the WF values of the individual ingredients composing each dish over the 4 weeks of the planned old menu and the 8 weeks of the planned revised one. Each meal was assessed, considering the recommended portion sizes for the primary school lunch menu specified in the guide to constructing the school canteen menu [25].

Food waste evaluation

Food waste (FW) was evaluated after the introduction of the new menu 21 times from October 2023 to May 2024. The evaluation was conducted in the canteen of the

primary school, with 16 classes including children aged from 6 to 11 years and teachers. The canteen serves up to 145 users.

Food waste was collected and measured at the canteen level, after meal distribution, without any direct interaction with or data collection from the children. No personal or individual data were recorded, and the aim was to evaluate the environmental impact and waste associated with the meals as served in real conditions.

The FW assessment was carried out by reference nutritionists, in collaboration with the school canteen staff. Meals were prepared in the school kitchen and served directly in the school canteen.

The served meal weight was calculated by multiplying the number of students and teachers present by the average serving of edible food (g). Food waste (kg) was obtained by weighing the food left on each plate (g) and not-served food (kg) categorizing it into first courses, second courses, and side dishes, excluding fruit and bread. The food was placed in bags and weighed separately for each course using a kitchen scale. After the weighing of each course, also the total waste was evaluated.

The percentage of food waste was computed as the ratio between the total edible plate waste (kg) and the total amount of prepared food (kg) referred to each course and as total.

Data analysis

Statistical analysis was performed using GraphPad Prism version 10.3.1 for MacOS, GraphPad Software, San Diego, California USA. The normality of data distribution was tested through the Shapiro–Wilk test. Descriptive statistics were run, and data were expressed as number and percentage (%) when appropriate, as

mean ± standard deviation (SD), if data distribution was normal, otherwise as median and minimum and maximum. Differences between groups were estimated using Student’s t-test and non-parametric mean comparison test, as appropriate and among groups using Anova test and non-parametric test, as appropriate. $p < 0.05$ was considered statistically significant.

Results

The baseline menu was updated in alignment with the National Guidelines for Collective School Catering, the guidelines for developing school canteen menus, and the Minimum Environmental Criteria [12, 14, 15, 25].

The new menu was designed with a framework of general categories, providing flexibility while ensuring a balanced dish selection. Indeed, for each category, a specific list of dishes is provided, with clear guidance to alternate between them rather than repeatedly choosing the same option. For example, cereals combined with legumes form one category, with options such as pasta with chickpeas or spelt with lentils. Similarly, seasonal vegetables are emphasized, with examples included.

One of the key changes in the revised menu was its broadening from a 4-week cycle to an 8-week cycle, ensuring greater variety according to the principles of the Mediterranean Diet. While the overall frequency of most food items remained consistent with the previous menu, there were notable adjustments: the frequency of legumes changed rising from less than once per week to 2 times per school week (5 days), while meat consumption was reduced from 1–2 times per school week to just once per school week, as outlined in Table 1.

Table 1 Comparison of meal frequencies with reference standards from Italian ministry of health

	Old menu	Revised menu	Reference standards from italian ministry of health
Fruit and vegetables	1/day	1/day	1/day
Cereals	1/day	1/day	1/day
Bread	1/day	1/day	1/day
Legumes	< 1/week	2/week	1–2/week
Potato	0–1/week	0–1/week	0–1/week
Meet	1–2/week	1/week	1–2/week
Fish	1/week	1/week	1–2/week
Eggs	1/week	1/week	1/week
Cheese	1/week	1/week	1/week
Processed meat	1/months is considered as 4 school week	0/months	up to 2/months
One-plate balanced meal	1/week	1/week	1/week

The week has to be intended as a school week (5 days)

Although the overall frequency of most food items remained largely unchanged from a quantitative point of view, changes were made in terms of quality.

In the baseline menu, the grain options were primarily limited to rice. Instead, the revised menu includes a more diverse selection, incorporating barley and spelt alongside rice. The consumption of white meat was reduced from twice a week to once a week, while red meat intake was adjusted from twice a week to three times per 8 school week. Processed meats, which were previously included, have been entirely removed from the menu.

Fish frequency remains the same, but the way it is prepared has been changed. Specifically, pre-breaded and pre-fried fish options have been replaced with either unbreaded fish or fish that is breaded in-house.

Legumes frequencies have been increased including them in the menu both as complete main courses and as components of first-course dishes (in this specific case the portion was considered as an half), such as pasta with lentil cream or spelt with chickpeas. This strategy was adopted to facilitate children’s gradual exposure to the flavor profile of legumes, promoting their acceptance and integration into dietary habits.

Desserts and cakes were reserved only for special occasions such as Christmas, Carnival, and Easter.

Table 2 highlights the most notable changes introduced in the revised menu.

Nutritional assessment

The nutritional assessment conducted on both the baseline menu and the revised menu revealed no significant differences in the majority of their nutritional values using unpaired t-test to compare normally distributed

variables and Mann–Whitney test to compare not normally distributed variables. However, a noteworthy exception was observed in the fiber content, which demonstrated a marked increase in the revised menu. Specifically, the average fiber content rose from 7.8 ± 0.6 g per meal in the baseline menu to 8.9 ± 1.2 g per meal in the updated version. A detailed comparison of the nutritional values between the baseline menu and the optimized menu is presented in Table 3.

Environmental impact assessment

The evaluation of both the carbon footprint and water footprint of the menu revealed a statistically significant difference between the baseline menu and the revised version in terms of carbon footprint but not for water footprint (unpaired t-test). Specifically, the carbon footprint showed a decrease from 5.2 ± 1.6 kg CO₂ equivalent per kilogram of meal in the baseline menu to 3.7 ± 0.9 kg CO₂ equivalent per kilogram of meal in the updated version ($p = 0.0489$; mean difference: -1.5 ; 95% CI -3.05 to -0.01). The water footprint was reduced from 5176.0 ± 738.9 L of water per kilogram of meal to 4608.0 ± 708.8 L per kilogram of meal ($p = 0.2254$; mean difference: -568.0 ; 95% CI -1548 to 411.6).

To better identify the main contributors to the overall environmental impact, each dish was individually assessed. This analysis showed that “meat” (in particular red-meat), “fish” and “dairy products” are the food categories with the highest carbon and water footprint. The comparative analysis of the new menu compared to its previous version reveals a remarkable increase in the variability of dishes thanks to the shifting towards 8 weeks-cycle instead of 4 weeks. The updated menu features a

Table 2 Main changes implemented in the transition from the old menu to the revised menu

	Old menu	Revised menu
Shift over weeks	4-week rotation	8-week rotation
Food group		
Legumes	1/week (autumn/winter) 1-2/week (spring/summer)	2/week (all seasons)
Peas	Served as seasoning for the first course or as a side dish	Served as seasoning for the first course or as a second course
Whole grains	Rice	Expanded variety: rice, barley, spelt
Meat	Avg. 2 times/week in both autumn–winter and spring–summer menus	About 1 time/week, favouring other second courses like legumes
Red meat	Avg. 2 times/week in autumn–winter; 1 time/week in spring–summer	3 times in 8 weeks (including unique dishes like lasagna)
Processed meats	1 time/week (e.g., cold cuts)	Eliminated processed meats
Fish	About 1 time/week	About 1 time/week with more variety (cod, plaice, cuttlefish)
Fish preparation	Included pre-breaded, pre-fried fish not prepared in-house	Removed pre-breaded, pre-fried fish not made by the kitchen
Cakes	2 times in 4 weeks (autumn–winter); 1 time/week (spring–summer)	Limited to special occasions, replacing fruit with simple local recipes

The week has to be intended as a school week (5 days) and months as 4 school week

Table 3 Nutritional assessment of the baseline menu and the revised menu

	Baseline menu	Revised menu	p	Difference between means	95% CI
Energy					
Kcal/meal	720.4 ± 32.3	707.9 ± 53.9	0.7040	− 12.5	− 89.36 to 64.32
Carbohydrates					
g/meal	105.4 ± 7.8	107.3 ± 13.6	0.8171	1.9	− 17.24 to 21.01
% Energy	58.5 ± 2.8	60.5 ± 3.3	0.3903	2.0	− 3.26 to 7.22
Sugars					
g/meal	14.0 ± 0.6	13.7 ± 0.9	0.6584	− 0.3	− 1.62 to 1.10
% Energy	7.8 ± 0.5	7.8 ± 0.8	0.9644	0.0	− 0.79 to 0.76
Proteins					
g/meal	28.2 ± 1.1	26.2 ± 1.5	0.0709	− 2.0	− 4.26 to 0.23
% Energy	15.7 ± 0.8	14.8 ± 0.8	0.1683	− 0.9	− 2.17 to 0.48
Animal					
g/meal	12.8 ± 1.8	10.2 ± 1.9	0.0879	− 2.6	− 5.75 to 0.53
%/total proteins	45.5 ± 6.7	39.5 ± 7.1	0.2651	− 6.0	− 17.95 to 5.95
Vegetable					
g/meal [‡]	14.2 (13.6–18.5)	14.9 (14.4–19.4)	0.2000 [^]		
%/total proteins	53.8 ± 7.1	60.5 ± 7.1	0.2283	6.7	− 5.56 to 19.06
Fat					
g/meal	21.6 ± 1.9	20.4 ± 1.1	0.3082	− 1.2	− 3.96 to 1.49
% Energy	16.1 ± 2.8	13.1 ± 2.9	0.1901	− 3.0	− 7.92 to 1.96
Saturated fat					
g/meal	4.2 ± 0.4	4.0 ± 0.6	0.5622	− 0.2	− 1.02 to 0.61
% Energy	5.2 ± 0.7	5.0 ± 0.9	0.8016	− 0.2	− 1.56 to 1.26
Fibers					
g/meal	7.8 ± 0.6	8.9 ± 1.2	0.1620	1.1	− 0.57 to 2.70
g/1000 kcal	10.9 ± 0.6	12.5 ± 0.8*	0.0160	1.6	0.44 to 2.88

Values reported as mean ± sd.; [‡] expressed as median (min–max) * Unpaired t-test * p < 0.05. [^] Mann–Whitney test. % Energy refers to the total meal energy content

reduced selection of meat-based offerings, particularly red meat and processed meat, which have been largely replaced by vegetable-based alternatives.

CF significant decrease is primarily attributed to the overall reduction in meat-based dishes coupled with a marked increase in plant-based alternatives while WF value reflects the increase in plant-based food that compensates the saving achieved from reducing meat.

Although in both the old and revised menus, the dishes with the highest footprint are still those based on meat, especially red meat (such as meatballs), the revised menu demonstrates how the newly introduced options have a significantly lower impact in terms of both CF and WF (Figs. 1 and 2).

Food waste evaluation

After the establishment of the revised menu, food waste was recorded 21 times from October 2023 to June 2024. The mean value of food waste percentage per course and the total are shown in Fig. 1. Side course

represents the one with the highest percentage of waste (31.4 ± 13.4)%, Fig. 3.

The evaluation process involved the quantification of the waste of each served dish, allowing to rank dishes from most to least discarded both in total (Fig. 4) and divided by courses (Fig. 5). The mean value for food waste in total is 27.3 ± 14.0%. Among the 63 dishes analyzed, a total of 29 were found to exceed the average food waste percentage of 27.4%. Within this subset of 29 dishes, spinach emerged as the item with the highest level of discrepancy observed followed by cabbage salad and stewed lentils.

This analysis examines food waste (FW) percentages in first, second, and side courses. A mean food waste percentage of 24.4 ± 10.0% was observed for the first course. Among the 22 dishes analyzed, 10 exhibited food waste percentages exceeding this mean. Notably, pasta with broccoli rabe demonstrated the highest waste, followed by pasta with lentil ragù and saffron rice. In contrast, the dishes with the lowest waste

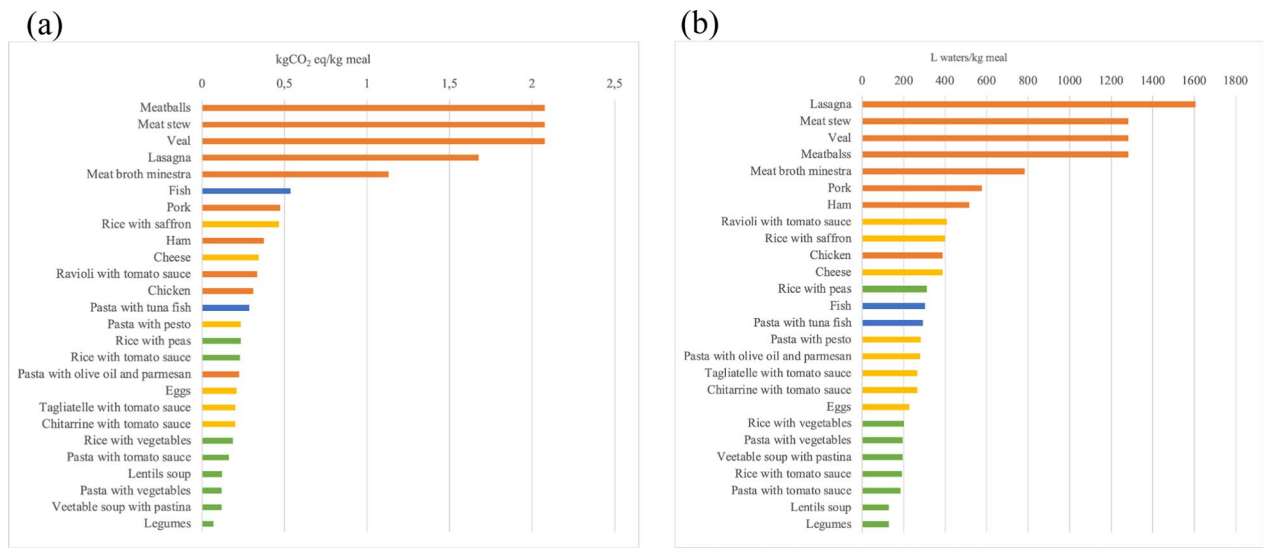


Fig. 1 Carbon footprint (kgCO₂ eq/kg meal) (a) and Water footprint (l waters/kg meal) (b) of each dish in the old menu categorized by type: meat-based in orange, fish-based in blue, vegetarian in yellow, and vegan in green

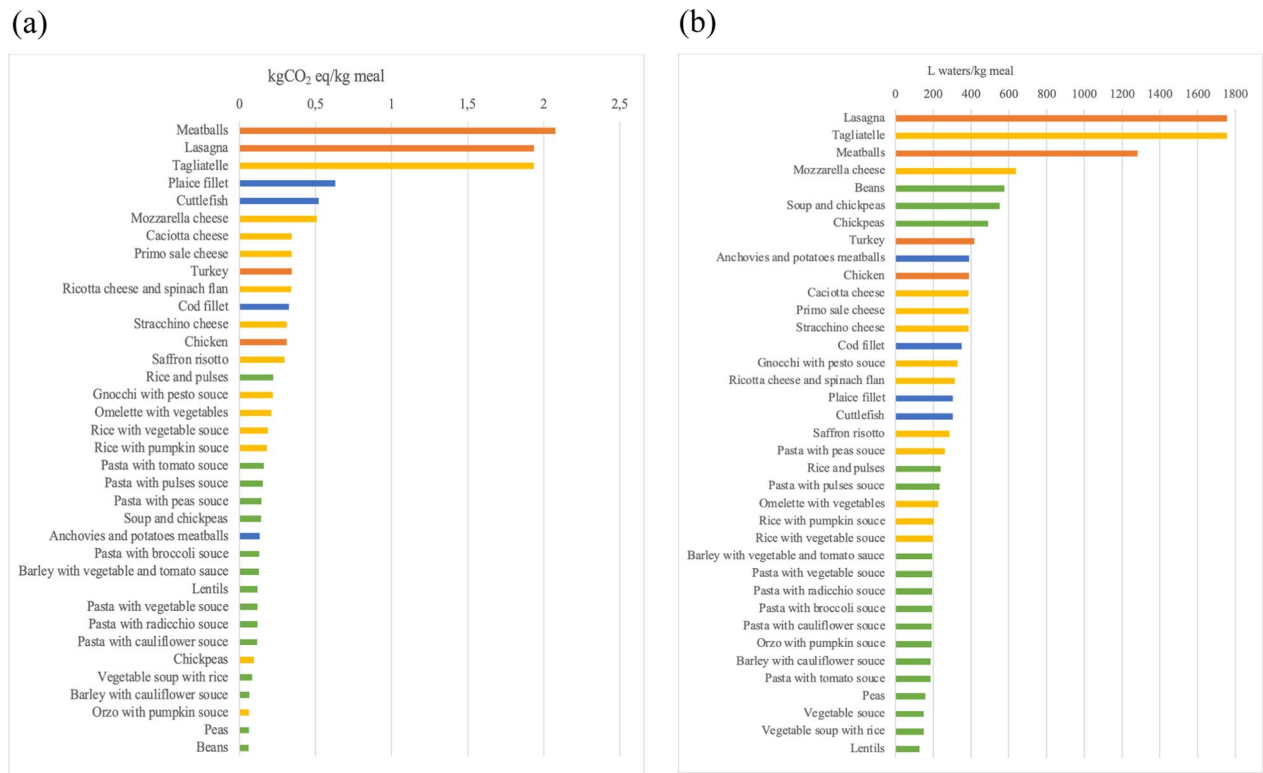


Fig. 2 Carbon footprint (kgCO₂ eq/kg meal) (a) and Water footprint (l waters/kg meal) (b) of each dish in the revised menu categorized by type: meat-based in orange, fish-based in blue, vegetarian in yellow, and vegan in green

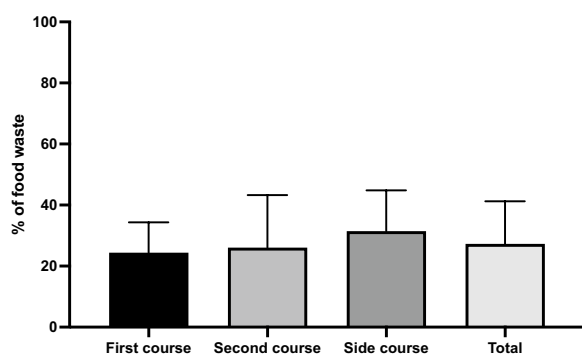


Fig. 3 Percentage of food waste, calculated as the total cooked weight in kilograms divided by the waste generated in kilograms, presented both for the overall total and for individual courses

included chickpea soup, ravioli, and pasta with broccoli (Fig. 5).

In the analysis of the second course, a similar pattern emerged, with 9 out of 22 dishes showing food waste percentages exceeding the mean of $26.0 \pm 17.2\%$. The most discarded items were those composed of stewed legumes and ricotta flan, while dishes centered around chicken exhibited the lowest levels of waste (Fig. 5).

For the side courses, a mean food waste percentage of $31.4 \pm 13.4\%$ was recorded, with 8 out of 22 dishes exhibiting food waste percentages surpassing this average. The data reveal considerable discrepancies in food waste among various side dishes. For instance, while spinach and raw fennel were among the most discarded, other side dishes, such as carrots, displayed significantly lower waste. This variability suggests that even the same dish, such as raw fennel, can exhibit divergent waste rates, indicating that factors such as preparation methods, presentation styles, and consumer preferences may significantly influence the extent of waste associated with specific side dishes (Fig. 5).

Discussion

In this case study, a school catering menu was updated with careful consideration of both nutritional quality and environmental sustainability. The revised menu aimed to balance the dietary needs of students while also incorporating sustainable practices, reflecting a comprehensive approach to health and sustainability in school meal planning.

Our primary aim was to update the school menu in alignment with the National Guidelines for Collective School Catering, the guidelines for developing school canteen menus, and the Minimum Environmental Criteria [12, 14, 15, 25].

The menu serves as an educational tool to guide children toward healthier eating habits, encouraging them

to try new foods, experiment with different recipes, and explore traditional dishes. The literature also highlights the importance of using school meals as a tool for nutritional education and education for sustainable consumption and practical learning activities, such as school gardens and cooking activities [11, 27–29].

It is widely recognized that children often face specific dietary challenges, such as limited intake of fruits and vegetables, excessive consumption of processed foods, and reluctance to try unfamiliar items. The WHO Childhood Obesity Surveillance Initiative (COSI), a surveillance initiative that involved 45 countries in 2020, in the COSI round 5 (2018–2020) reported that overall, 43% of children aged 6.9 years consumed fresh fruit daily and only 34% ate vegetables daily [3]. According to the 2023 Italian National report (OKkio alla Salute), fruit, vegetable, and legume consumption among Italian children remains a concern. According to the survey, 25.9% of children eat fruit and vegetables less than once a day and 37% of children consume legumes less than once a week, in addition, almost 10% of the population skip breakfast, and over half consume excessive morning snacks. Nearly 25% drink sugary beverages daily [2].

Also, a recent study conducted in 2023 by Di Nucci et al. in Italy highlighted a low adherence to the Mediterranean Diet among children. Using the KIDMED questionnaire, the study revealed significant deficiencies in the consumption of fruits and vegetables. Additionally, 53.8% of the children reported eating pulses less than once per week. In contrast, there was a high intake of sweets and processed foods, further indicating poor adherence to the core principles of the Mediterranean Diet [30].

These highlight the critical importance of engaging and supporting children in making healthier dietary choices stressing how it is essential to support their integration of diverse, nutrient-rich foods into their daily meals, while also emphasizing the importance of seasonality and sustainability in food choices.

Schools represent a critical environment for influencing the dietary habits of children and adolescents, as students spend a significant portion of their day at school and frequently consume at least one meal on-site. The integration of health and nutrition within the educational framework is increasingly acknowledged as essential, with schools playing a pivotal role in fostering healthy eating behaviors through multifaceted interventions, including modifications to the physical setting, nutrition education, and comprehensive food policies. These initiatives not only aim to ensure that students have access to safe, nutritious, and diverse meals, but also seek to introduce sustainable dietary practices that can

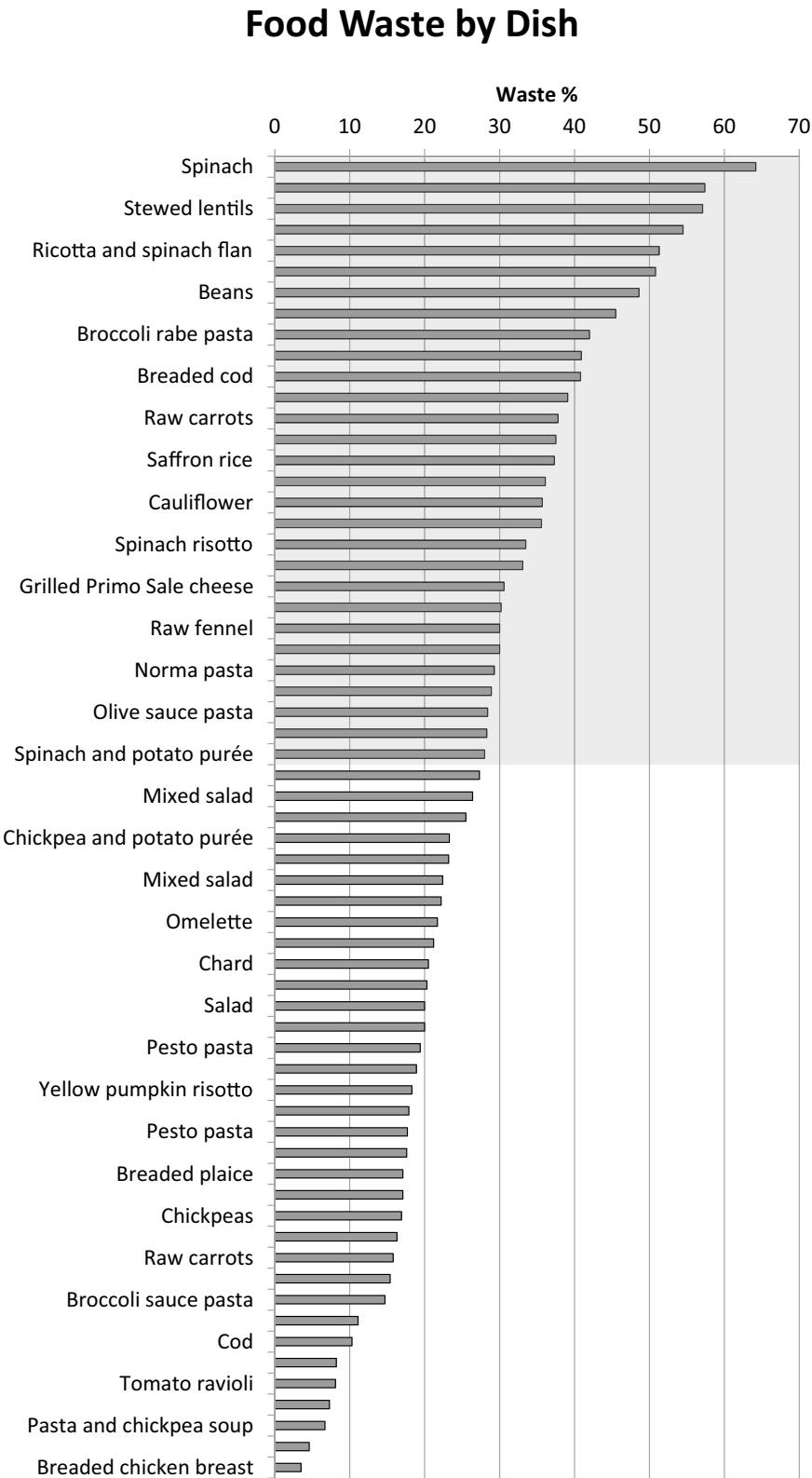


Fig. 4 Food waste per each evaluated dish, listed from the most discarded items to the least. Shaded areas represent dishes presenting a food waste percentage above the average value. When the same dish appears more than once, it is because it was served on multiple days during the waste evaluation period

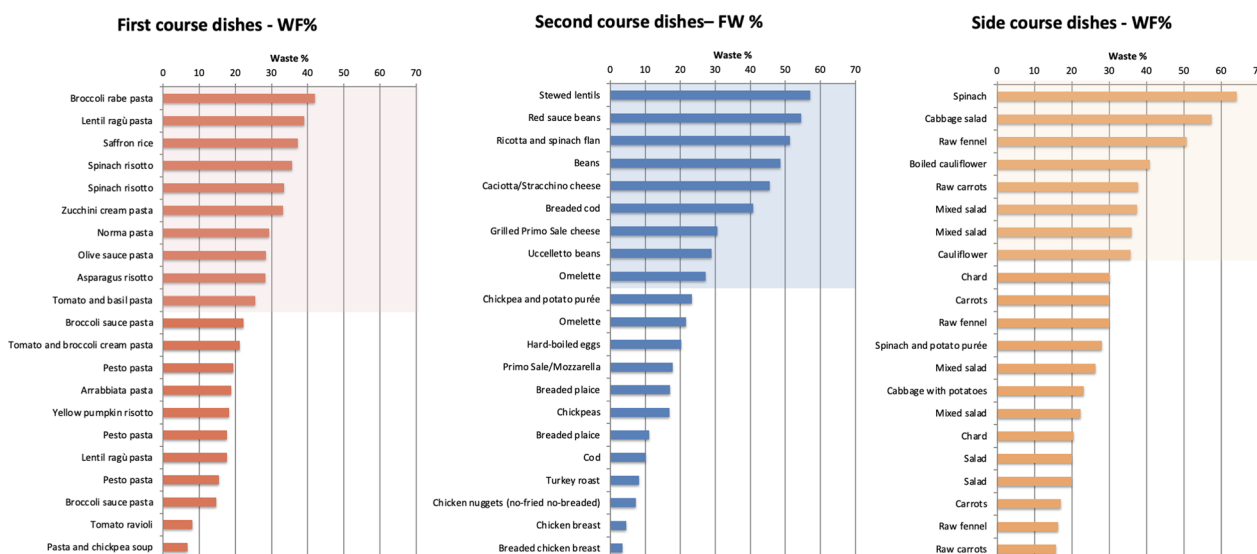


Fig. 5 Food waste for each evaluated dish categorized by course (red: first course dishes; blue: second course dishes; orange: side course dishes), listed from the most to the least discarded items. Shaded areas represent dishes presenting a food waste percentage above the average value for their respective category. When the same dish appears more than once, it is because it was served on multiple days during the waste evaluation period

positively influence their households and local food systems [31–33].

In this case study, the menu was initially changed in terms of weeks, indeed a shift from 4-week cycle to an 8-week cycle was carried out in order to increase the variety of foods.

One of the principles of the Mediterranean diet is the advice to vary food choices to ensure differences in appearance and taste as well as macronutrients, micronutrients and phytochemical contents [13]. In addition, more variability in food choice is also important in terms of food neophobia. As reported by Di Nucci and colleagues, food neophobia can represent an obstacle to achieving a nutritious, balanced diet, particularly in children. Indeed, they underlined that food neophobia is a key factor contributing to a lower adherence to the Mediterranean Diet with a negative impact on the intake of foods, such as fruits, vegetables, legumes, and nuts, while promoting a higher consumption of sweets and fast food. Although food neophobia is a normal developmental phase that typically occurs between 2 and 6 years of age, there are factors able to contribute such as an unhealthy home food environment, poor parental eating habits, and ineffective feeding practices. Thus, supporting children to gradually diversify their food preferences is critical for fostering healthier eating habits and improving long-term well-being [30].

According to this, the optimized menu aims to increase nutritional knowledge and ameliorate the food habits among children. Changes and updated of the menu

were carried out in order to increase legumes consumption. The only grain cereal on the baseline menu was rice, instead in the new one rice, barley and spelt are present. The meat was proposed twice a school week, in the revised one its consumption is reduced to once a school week and replaced by other protein sources, such as pulses. In the revised menu, red meat was reduced to 3 times in 8 school weeks and the processed meat is eliminated. Fish is maintained once a school week, with the advice to vary the typology and the elimination of pre-cooked fish. Early experience with a variety of novel foods increase children's willingness to try new foods [28], thus the implementation of diverse dishes based on the Mediterranean Diet could facilitate a greater acceptance of various flavors and ingredients among children, thereby promoting healthier eating habits.

To enhance children's dietary patterns and nutritional intake, explore a wider range of foods is important. In this context, the introduction of new tastes and the increasing of plant-based foods are key actions to be considered in the development of school canteen's menu [29, 34]. The inclusion of different plant-based foods and the addition of new recipes can enhance children's acceptance of healthier options and, at the same time, support environmental sustainability [29, 34].

Our updated menu also aimed to widen children's taste experiences by not only adding plant-based dishes but also increasing the variety of food options. This approach aligns with findings suggesting that repeated exposure to a diverse set of foods can significantly contribute to

dietary acceptance and long-term behavioral change [28, 29, 34].

The school menu has a central role in the context of the promotion of environmental awareness and sustainability. Indeed, schools can reduce the ecological footprint by including plant-based options, local sourcing, and minimizing food waste while providing healthy meals. The selection of dishes has a crucial role in defining the environmental impact of dietary patterns. Well-planned menus can significantly reduce greenhouse gas emissions compared to traditional options [5, 10, 35, 36].

In our study, the update of the Camerino primary school menu, aligned with Italian nutritional guidelines and focused on increasing plant-based dishes. In this context, the carbon footprint (CF) per meal decreased from 5.2 to 3.7 kg CO_{2eq}, mainly due to the reduction of red and processed meat dishes and the inclusion of plant-based alternatives such as legumes and cereals.

This results are in line with previous studies such as the menu optimization modelling study conducted in the municipality of Rome through binary linear programming model which projected up to a 40% reduction in emissions for optimized omnivorous menu when compared to standard menus [37]. This finding underscores the potential of specific menu planning to enhance sustainability in school meal programs.

Our results demonstrates that it is possible to reach substantial reductions in the carbon footprint of school menus considering also cultural dietary preferences.

In line with Volanti et al., who assess the environmental impact of meals served in school canteens using a life cycle perspective, our results confirm that animal-based dishes, especially red-meat and processed meats, are the most carbon-intensive options while meals based on legumes, grains and seasonable vegetables show lower impacts. In Volanti's study, dishes containing fish and meat were the most carbon-intensive, with emissions ranging from a minimum of 3 g CO_{2eq} to a maximum of 24 g CO_{2eq} [38] reinforcing our findings regarding the importance of food type and composition.

In addition, our findings align also with other European experiences. In Sweden, Eustachio Colombo and colleague reported that optimized school meals could reduce greenhouse gas emissions by up to 40%, without compromising nutritional adequacy [39]. In the UK, a study evaluates the GHG emissions of school nursery lunches and confirmed that meat especially red meat-based meals and meals containing any meat/fish were significantly higher in GHG emissions than vegetarian meals [40]. Similarly, Martinez and colleagues, analysing six spring and six fall school menu for 7–12 years old children underlined that Menus including non-meat protein sources, such as legumes, exhibit the lowest Nitrogen

pollution [41]. Beyond Europe, Kluczkowski and colleagues conducted a dual environmental and nutritional evaluation of school meals in Brazil and demonstrate the potential of school meals as both nutritional and environmental interventions [42].

These results may be particularly relevant for Mediterranean countries, where traditional dietary patterns are similar. School meals should be developed based on MD specifically designed to meet both nutritional needs and environmental sustainability goals taking into account also that successful implementation will require careful consideration of local procurement practices, student food preferences, and education system frameworks.

However, our findings also highlight the aspect that some plant-based foods can have a different environmental impact considering the different indicators. For example, some vegetables can have a relatively high water footprint. This has been highlighted by Martínez-García and colleagues which showed that fruits were among the most water-demanding food groups in Spanish school meals [43]. The presence of such items among the most frequently discarded dishes highlights the importance of considering water use in environmental assessments.

While efforts to reduce the carbon footprint and water footprint of food production are crucial for sustainability, to monitor and minimize food waste is also important. The importance of focusing the attention of food waste in canteens and institutional catering is underlined by the Italian CAM [15].

CAM recommend continuous monitoring to quantify the amount of waste generated, which is essential for identifying inefficiencies in food management and consumption patterns. Thus, the evaluation of food waste must be a central part of sustainability strategies.

In this study, an analysis of food waste was conducted after the introduction of the revised menu. The results revealed an average waste of 24.4% for first courses, 26.0% for second courses, and 31.4% for side dishes. These results are slightly lower than those reported in previous studies conducted in Italian school settings. For instance, research conducted in Bari found waste rates of 35% for first courses, 32% for second courses, and 55.4% for side dishes [44] while a study in Bologna reported 29.6%, 38.3%, and 57.7%, respectively [45].

Despite the slightly improved waste levels, a consistent trend across all studies seems to persist, with the highest levels of waste attributed to side dishes, which predominantly consist of vegetables. This is a known issue, as vegetables are one of the most challenging food groups for children, as also highlighted in the 2023 “Okkio alla Salute” report [2].

These findings point to a fundamental challenge in sustainable menu design: the inclusion of nutritionally

and environmentally desirable foods is insufficient if not accompanied by strategies to improve acceptance. As highlighted in the DELICIOUS project and related behavioral studies, student involvement in tailored activities is essential to increase Mediterranean diet adherence and to reduce enhance dietary sustainability [8].

This study presents limitations that should be acknowledged. Firstly, it lacks a pre-post design to assess food waste. This was due to the fact that the project began with the implementation of the new menu thus not enabling the preventive collection of baseline data. This lack limits a direct comparison and leads to no possibility to study the intervention's effectiveness. However, this study can be considered a starting point for future research incorporates both pre- and post-intervention assessments, enabling a more robust evaluation of the impact of sustainable menu revisions. Our findings results of great importance as they represent a picture of current waste patterns offering direction for further research. Notably, plant-based dishes showed the highest waste rate, a finding consistent with previous studies [46, 47]. This could be related to several factor as the level of satisfaction of respect to the sensory characteristics of dishes [48, 49].

This underline the importance of incorporate specific strategies aiming to improve dishes' taste and presentation, involve children in the co-creation of recipes, but also implement nudging strategies [50, 51].

Secondly, the study did not investigate the nutritional knowledge, food preferences, or eating behaviors of the participating children, which could have provided further insight into the determinants of food acceptance and waste. Future study will include both nutritional knowledge and food habits assessment and nutritional educational intervention. Another limitation of this study is the lack of data on micronutrient intakes that is particularly relevant in the context of plant-based diet. However, accurately quantifying the micronutrient content of the proposed menu is challenging, as existing nutritional analysis software presents limitations in terms of precision for micronutrient estimates. Additionally, the effects of food preparation, cooking methods, and potential nutrient losses are not consistently accounted for. This limitation is also reflected in the Italian school meal guidelines, which provide quantitative recommendations only for macronutrients [12]. Nevertheless, there are emerging evidence that underlines the possible deficiencies (for example in Vitamine B12 and iron) that could be encountered in plant-based dietary approaches. Desmond and colleagues found that vegan children aged 5–10 years exhibited a healthier cardiovascular risk profile but a higher prevalence of micronutrient deficiencies compared to their omnivorous peers while vegetarian children showed fewer nutritional inadequacies [52]. In

addition, a recent systematic review included that both plant-based and omnivorous children are at risk of inadequate intakes of essential nutrients. Across all diet types, vitamin D, calcium, and iodine were frequently suboptimal. Children following plant-based diets were more likely to have insufficient intakes of iron, zinc, vitamin B12, and omega-3 fatty acids, while omnivorous children showed risks of inadequate folate and vitamin E intake [53]. This underlining the importance of ensuring a balanced and adequate nutritional approach in each situation. Moreover, it is important also to note that in this study the plant-based meals are not part of a fully vegetarian or vegan approach. The frequency of plant-based dishes served at school increased but children were not restricted from consuming animal products in their daily diet. Taking into account these considerations, it is important to underline the need for future studies to assess both micronutrient intake and status, regardless of dietary pattern, to ensure a comprehensive understanding of nutritional adequacy in children. To address this gap, it is important to conduct studies to comprehensively evaluate macro- and micro-nutrients intakes and micronutrient status.

In addition, the environmental impact evaluation was based on data from the Barilla Center for Food and Nutrition (BCFN) database, which, while relevant to the Italian context, carries intrinsic limitations described in methods section. One key limitation is the use of standardized values that reflect average European production systems and do not distinguish between local and non-local sourcing. This enables comparability across meals and food categories, but it may not fully capture geographic variability. As highlighted by Petersson and colleagues, future research could improve precision by integrating regionally data, where available, and re-analyzing primary inventory data by geographic grouping [26].

Despite these limitations, the study offers an important contribution to the evidence base on sustainable school meal programs, providing integrated insights into both environmental impacts and behavioral outcomes. It offers a valuable foundation for future multi-dimensional assessments and policy-oriented interventions aiming to improve the sustainability and effectiveness of institutional food services.

Conclusion

This study highlights that school meal interventions integrating menu optimization and sustainability criteria can reduce environmental impacts while maintaining nutritional quality. To enhance their effectiveness, school nutrition programs should include structured food education interventions, pre-implementation acceptability

testing, and systematic food waste monitoring. Policy structures should also highlight the use of seasonal, locally sourced, and culturally suitable foods to support sustainable and acceptable dietary transitions in school settings. The systematic monitoring of food waste is of fundamental importance with in parallel the presence of educational campaign focused also on food acceptance through participatory activities and taste education in particular for vegetables. Also, it is important that the national meal guidelines formally incorporate environmental sustainability criteria. All these measures are essential to align school food systems with both nutritional goals and sustainability policies.

Abbreviations

CAM	Minimum Environmental Criteria (Criteri Ambientali Minimi)
CF	Carbon footprint
FW	Food waste
GHG	Greenhouse gas
MD	Mediterranean diet
NCDs	Non-communicable diseases
WF	Water footprint
WHO	World Health Organization

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Author contributions

Conceptualization VP, LB, GV; methodology, VP, GV, DG; Investigation, DC, SZ, LM; Data analysis, GV, DG; Writing-Original draft preparation, GV, DG, LM; Writing-Review and Editing SR, SP, SV, Editing: LB, SP. All authors read and approved the final manuscript.

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The authors declare that they have no competing interests.

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